NCERT

National Cyber Emergency Response Team

DIGITAL FORENSICS LABORATORY — GROUP PROJECT REPORT

Evidence Management System (with Chain of Custody)

Phase 2 Report — Core Operational Modules

50% Project Completion

Group:	Group 2
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Instructor:	Sir Umar Javaid
Course:	Digital Forensics Internship — Project Completion Phase 2
Platform:	Next.js 14 · PostgreSQL · Prisma · NextAuth · SHA-256
Submission:	July 2026
Progress:	50% Complete (Core Modules)

Academic Integrity Notice: All design, implementation, and verification activities were performed on an isolated, instructor-authorized laboratory / development system for the NCERT Digital Forensics Internship. No production case systems or third-party operational evidence stores were accessed at any stage of this project.

Contents

Abstract

This report documents **Phase 2** of Group 2's NCERT Digital Forensics Internship project *Evidence Management System (with Chain of Custody)*, supervised by **Sir Umar Javaid**. Building on Phase-1 research and architecture, Phase 2 implements the first operational half of the platform and advances overall completion from 25% to **50%**. Delivered modules include Credentials authentication with role claims, evidence registration with unique IDs and ingest-time SHA-256 hashing, custody event recording (seizure, transfer, examination, return), a role-aware dashboard shell, and baseline account settings. Integrity mismatch workflows, printable PDF reports, full audit analytics, and admin user management remain scheduled for Phase 3.

1 Introduction

Phase 1 produced requirements and a design blueprint. Phase 2 converts that blueprint into running software that a trainee can sign into and use to register exhibits and move them through a custody ledger. The emphasis is forensic correctness of the core loop—**hash at ingest, authenticate actors, append custody events**—rather than cosmetic completeness.

1.1 Objectives

1. Stand up the Next.js application with PostgreSQL, Prisma, and migrations.
2. Implement NextAuth login with custodian / examiner / supervisor / admin roles.
3. Deliver evidence registration with unique IDs and server-side SHA-256.
4. Implement custody events capturing handler, timestamp, location, and reason.
5. Provide a dashboard shell with module navigation gated by permissions.
6. Seed demo accounts for instructor walkthroughs.
7. Verify the Phase-2 modules with build checks and guided manual tests.

1.2 Scope

In scope: authentication, evidence CRUD/list/detail (registration focus), custody logging, dashboard overview, and settings password/profile basics. Out of scope for Phase 2 (reserved for Phase 3): integrity re-hash queue and supervisor resolution, custody PDF generation, full audit trail UI/export, admin user lifecycle screens, and marketing/landing polish.

1.3 Tools and Environment

Component	Details
Framework	Next.js 14 (App Router) + TypeScript strict
UI	Tailwind CSS + shadcn/ui primitives
ORM / DB	Prisma 5 + PostgreSQL (Docker Compose)
Auth	NextAuth.js v4 — Credentials provider, JWT sessions
Hashing	Node.js crypto.createHash('sha256')
Dev tooling	ESLint, tsc --noEmit, npm scripts
Seed accounts	admin / supervisor / examiner / custodian demos

Table 1: Tools and environment used during Phase 2 implementation.

2 Continuity from Phase 1

Every Phase-2 module traces to a Phase-1 requirement ID. FR-01–FR-05 and FR-09 (partial) are the primary targets. The RBAC matrix from Phase 1 is enforced through middleware route permissions and server-action `requirePermission()` guards so that hiding a menu item is never the only control.

Req.	Module delivered in Phase 2	Status
FR-01/02/03	Evidence registration + SHA-256 ingest	Implemented
FR-04/05	Custody event ledger + evidence timeline	Implemented
FR-09	Login + role-gated navigation / actions	Implemented (core)
NFR-01	Server-side permission checks on mutations	Implemented
FR-06/07/08/10	Integrity, PDF, audit UI, admin users	Phase 3

Table 2: Traceability from Phase-1 requirements to Phase-2 modules.

3 Methodology — Implementation Approach

- **Schema-first:** Prisma models for User, EvidenceItem, CustodyEvent; migrate; seed.
- **Secure-by-default routes:** middleware maps paths to permissions; pages re-check `can()`.
- **Server actions:** create evidence / log custody only after authz; hash computed on server.
- **Demo realism:** seed users with Password123! for instructor review without exposing production secrets.
- **Incremental verification:** typecheck + lint + smoke of routes after each module slice.

4 Module Implementation

4.1 Authentication and Session Security

Users authenticate with email/password. Passwords are stored as bcrypt hashes. Successful login issues a JWT session embedding user id, name, email, and role. Unauthenticated requests to dashboard routes redirect to `/login`. Role pills in the shell surface the active duty function during demos.

4.2 Evidence Registration with SHA-256

Examiners (and permitted roles) register exhibits via `/evidence/new`. The system allocates a unique human-readable ID, stores intake fields, and computes SHA-256 over the uploaded buffer (or accepts a double-entered external hash). The digest is persisted as the immutable baseline for later handoff checks.

Field	Purpose
Evidence ID	Unique registry key (EVD-YYYY-NNNN pattern)
Title / type / case reference	Inventory and case linkage
Intake location & notes	Context for later custody reporting
SHA-256 digest	Integrity baseline for handoff re-hash
Current custodian	Inventory ownership pointer
Status	e.g., IN_CUSTODY after successful registration

Table 3: Evidence registration fields captured at ingest.

4.3 Custody Event Ledger

Each movement is recorded as a custody event with type (SEIZURE, TRANSFER, EXAMINATION, RETURN), acting user, timestamp, location, and mandatory reason. The evidence detail page presents a timeline; the custody module provides a ledger-oriented view. Events are append-only: correcting history requires a new compensating event, never silent edits—matching forensic CoC expectations from Phase 1 research.

Event type	Typical actor	Effect on inventory
SEIZURE	Custodian / examiner	Initial intake into system custody
TRANSFER	Current custodian	Updates current custodian pointer
EXAMINATION	Examiner	Marks under-examination workflow
RETURN	Examiner / custodian	Returns item to holding custody

Table 4: Custody event types implemented in Phase 2.

4.4 Dashboard and Navigation Shell

The authenticated shell provides sidebar navigation, breadcrumbs, and overview statistics (counts of evidence, custody activity). Module accents differentiate Evidence (emerald) and Custody (amber) for rapid orientation during instructor demos. Routes the user's role cannot access are hidden and server-denied.

4.5 Account Settings Foundation

Users can view profile attributes and change their own password under `/settings`. This supports safe demo hygiene (rotating the seeded password on shared lab machines) without waiting for the Phase-3 admin console.

5 Database Migrations and Seed Data

Prisma migrations create the relational schema. Seed scripts insert Group-demo users spanning all four roles so Sir Umar Javaid and peer reviewers can exercise RBAC without manual user creation.

Role	Name	Email
ADMIN	Muhammad Ozair	admin@ems.local
SUPERVISOR	Imran Qureshi	supervisor@ems.local
EXAMINER	Sara Malik	examiner1@ems.local
CUSTODIAN	Nadia Hussain	custodian1@ems.local

Table 5: Seeded demo accounts (password for all: Password123!).

6 Verification Performed in Phase 2

#	Check	Result
1	PostgreSQL up; Prisma migrate + seed	Pass
2	Login as examiner / custodian / supervisor	Pass
3	Register evidence; SHA-256 stored	Pass
4	Log TRANSFER with reason; timeline updates	Pass
5	Custodian cannot open admin-only paths	Pass
6	TypeScript check (tsc --noEmit)	Pass
7	Production build of Phase-2 codebase	Pass

Table 6: Phase-2 verification checklist.

7 Progress Accounting (50%)

Workstream	Phase 1	Phase 2	Remaining (Phase 3)
Research & design	Done	—	—
Auth + RBAC core	Designed	Done	Admin UI hardening
Evidence + SHA-256	Specified	Done	List polish / filters
Custody ledger	Specified	Done	Velocity analytics
Integrity re-hash	Specified	Not yet	Full module
PDF custody report	Specified	Not yet	Full module
Audit trail UI	Specified	Partial hooks	Full module + export
Overall project	25%	50%	50%

Table 7: Cumulative completion after Phase 2.

8 Results Summary

Group 2 now has a working laboratory application that satisfies the first two bullets of the official brief (registration with hashes; custody event recording) and partially satisfies RBAC. The remaining brief items—handoff re-hash with printable reports, and full audit depth—are explicitly reserved for Phase 3 so that progress claims stay honest.

9 Conclusion

Phase 2 successfully advanced the Evidence Management System from design to an operable 50% system. Authentication, SHA-256 ingest, and the custody ledger form the forensic spine of the product. Phase 3 will complete the integrity verification loop, custody PDF packaging, audit trail, administration, and end-to-end verification against the Project 1 brief.

References

[1] Group 2 — Phase 1 Report: Evidence Management System (Research & Foundation), NCERT Internship.

[2] NCERT Project 1 brief: Evidence Management System (with Chain of Custody).

[3] Next.js Documentation — App Router & Server Actions.

[4] NextAuth.js Documentation — Credentials provider.

[5] Prisma Documentation — Schema, Migrate, Seed.

[6] NIST FIPS 180-4 — Secure Hash Standard (SHA-256).

[7] B. Carrier, *File System Forensic Analysis*. Addison-Wesley.

[8] Group 2 Week 3 NCERT laboratory report — formatting reference.

A Module File Inventory

Area	Path / artefact
Login	src/app/login/page.tsx
Dashboard	src/app/(dashboard)/dashboard/page.tsx
Evidence list / new / detail	src/app/(dashboard)/evidence/**
Custody	src/app/(dashboard)/custody/page.tsx
Settings	src/app/(dashboard)/settings/page.tsx
Actions	src/actions/evidence.ts, custody.ts, account.ts
Auth / RBAC	src/lib/auth*, src/lib/permissions*, middleware
Schema	prisma/schema.prisma + migrations + seed

Table 8: Representative Phase-2 source locations.